Perturbative QCD at the LHC

Rene Poncelet

Various Faces of QCD
16th to 18th October 2026

THE HENRYK NIEWODNICZAŃSKI
INSTITUTE OF NUCLEAR PHYSICS
POLISH ACADEMY OF SCIENCES

Funded by
the European Union

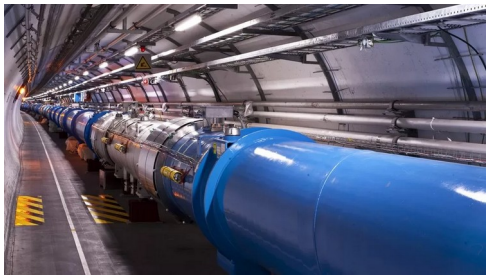
European Research Council
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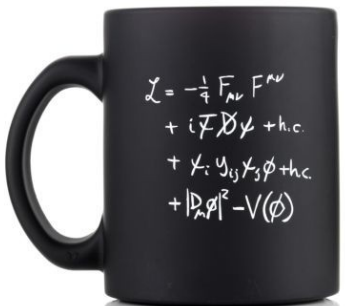
What are the fundamental building blocks of matter?

Scattering experiments

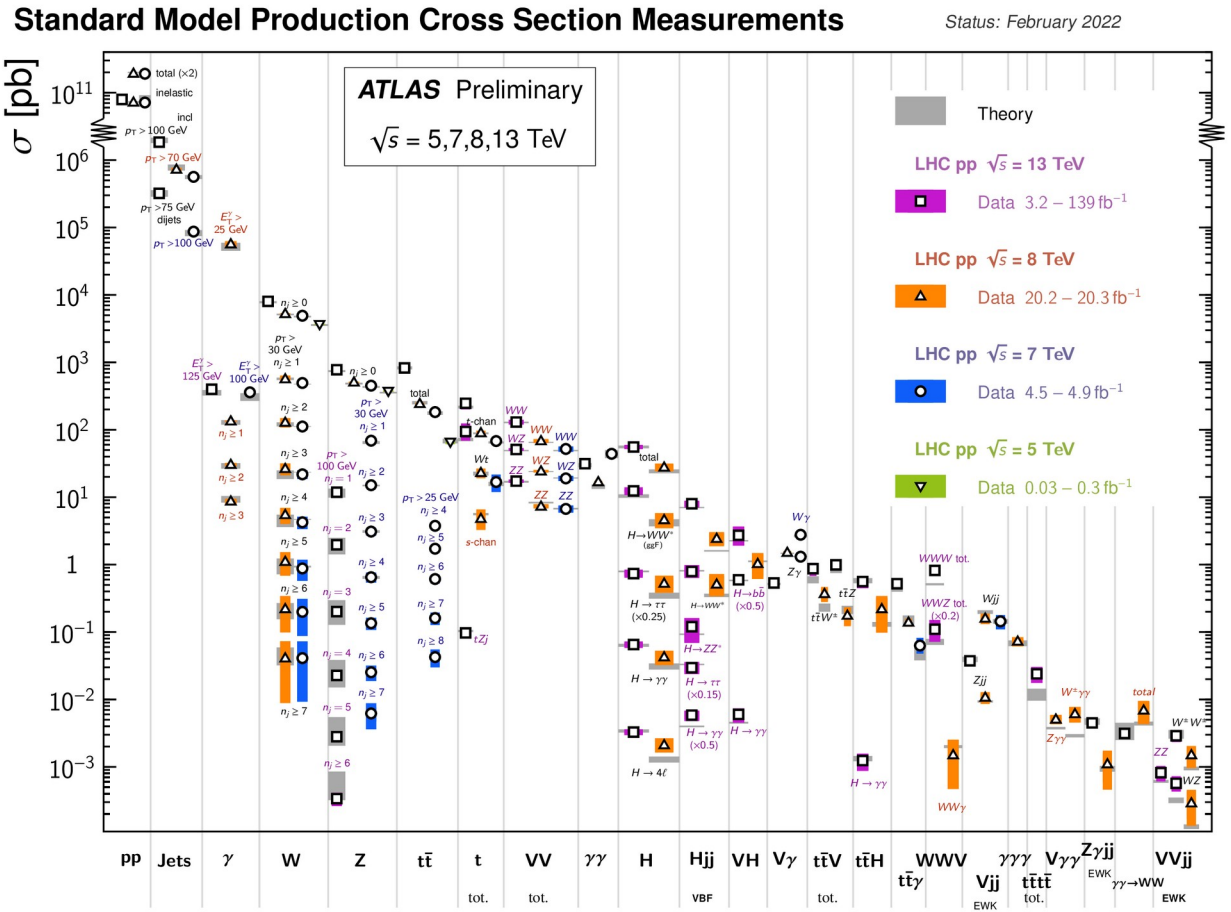
Large Hadron Collider (LHC)



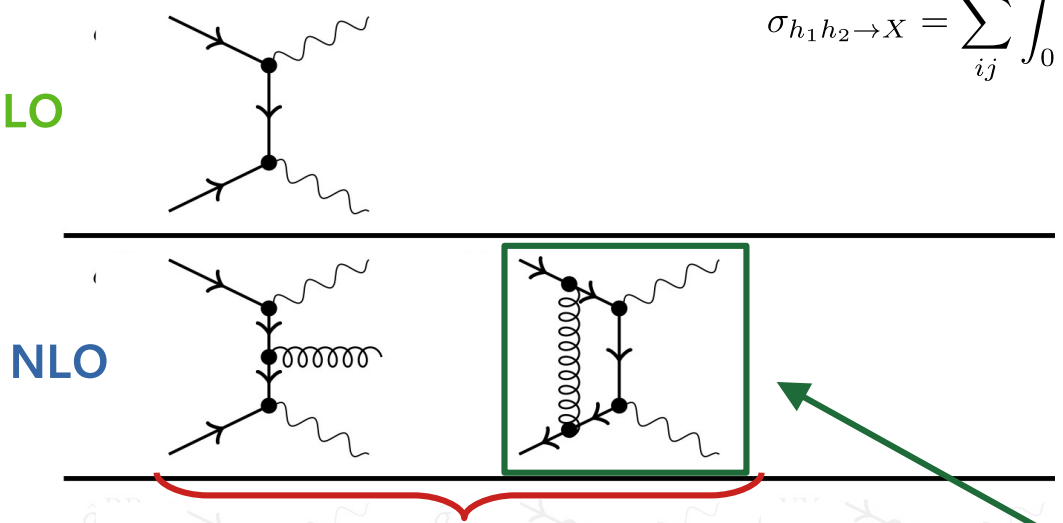
Credit: CERN



Theory/ Standard Model



Fixed-order QCD in collinear factorization



$$\sigma_{h_1 h_2 \to X} = \sum_{ij} \int_0^1 \int_0^1 \mathrm{d}x_1 \mathrm{d}x_2 \phi_{i,h_1}(x_1, \mu_F^2) \phi_{j,h_2}(x_2, \mu_F^2) \hat{\sigma}_{ij \to X}(\alpha_s(\mu_R^2), \mu_R^2, \mu_F^2)$$

partonic cross section

$$\hat{\sigma}_{ab \to X} = \hat{\sigma}^{(0)}_{ab \to X} + \hat{\sigma}^{(1)}_{ab \to X} + \hat{\sigma}^{(2)}_{ab \to X} + \mathcal{O}(\alpha_s^3)$$

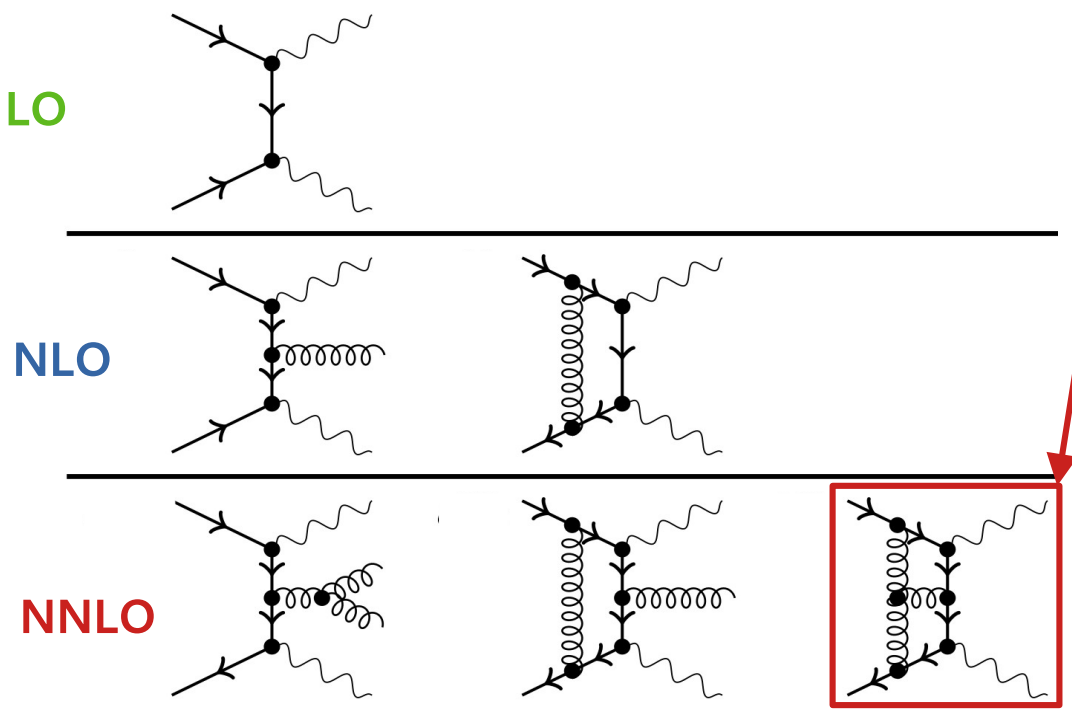
$$\hat{\sigma}^{(i)} \ni \frac{1}{\mathcal{F}} \int |\mathcal{M}_n|^2 \mathrm{d}\Phi_n$$

IR-finite cross section

Automated fixed-order @ NLO

- **oneloop-matrix elements**
  - Openloops, Recola, Helac, MadGraph, GoSam
- **local QCD/EW subtraction**
  - FKS: MadGraph5_aMC@NLO, Powheg
  - CS dipoles: Sherpa, Herwig

NNLO QCD challenges: two-loop amplitudes



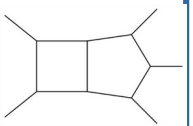
How to compute
multi-scale two-loop QCD amplitudes?

→ **fast growing complexity**:
rational coef. and **special functions**

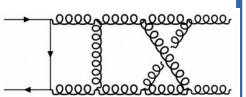
→ deeper understanding of the analytical properties

→ refinement of computational tools

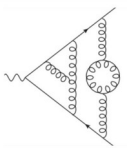
Two-loop 5-point
[Abreu, Agarwal, Badger, Buccioni, Chawdhry, Chicherin,Czakon, de Laurenties, Febres-Cordero, Gambuti, Gehrmann, Henn, Lo Presti, Manteuffel, Ma, Mitov, Page, Peraro, Poncelet, Schabiner Sotnikov, Tancredi, Zhang,...]



Three-loop 4-point
[Bargiela, Dobadilla, Canko, Caola, Jakubcik, Gambuti, Gehrmann, Henn, Lim, Mella, Mistleberger, Wasser, Manteuffel, Syrrakow, Smirnov, Trancredi, ...]



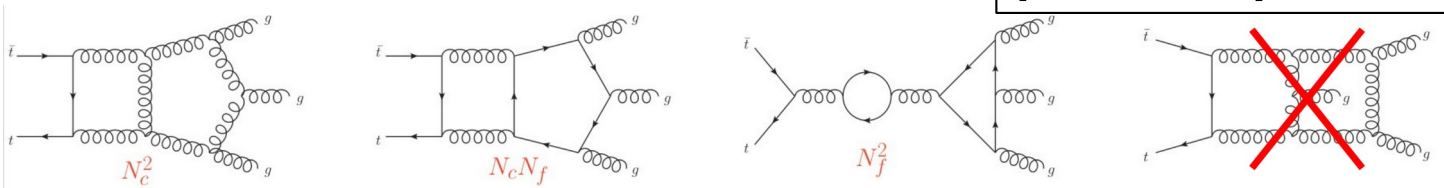
Four-loop 3-point
[Henn, Lee, Manteuffel, Schabinger, Smirnov, Steinhauser,...]



Two-loop virtual amplitudes $t\bar{t}$ + jet production

Sample diagrams

Double virtual QCD corrections to $t\bar{t}$+jet production at the LHC, Badger, Becchetti, Brancaccio, Czakon, Hartanto, **Poncelet**, Zoia [arxiv:2511.11424]



Colour & spin structures

$$\mathcal{M}^{(L)}\left(1_{\bar{t}},2_t,3_g,4_g,5_g\right) = \sqrt{2}\,\bar{g}_s^3\left[(4\pi)^\epsilon e^{-\epsilon\gamma_E}\frac{\bar{\alpha}_s}{4\pi}\right]^L \times \sum_{\sigma\in Z_3}\left(t^{a_{\sigma(3)}}t^{a_{\sigma(4)}}t^{a_{\sigma(5)}}\right)_{i_2}^{\bar{i}_1}\,\mathcal{A}_g^{(L)}\left(1_{\bar{t}},2_t,\sigma(3)_g,\sigma(4)_g,\sigma(5)_g\right)$$

Retaining full spin information
→ allows to incorporate spin-correlations in NWA decays

$$\mathcal{A}_x^{(L)h_3h_4h_5} = \sum_{i=1}^{4}\Psi_i\,\mathcal{G}_{x;i}^{(L)h_3h_4h_5}$$

Leading colour expansion of partial amplitudes:

$$\mathcal{F}_x^{(1)} = N_c\,F_x^{(1),N_c} + n_f\,F_x^{(1),n_f}\,,$$
$$\mathcal{F}_x^{(2)} = N_c^2\,F_x^{(2),N_c^2} + N_c n_f\,F_x^{(2),N_c n_f} + n_f^2\,F_x^{(2),n_f^2}\,.$$

Partial amplitudes: masters expressed in basis functions

$$\tilde{\mathcal{F}}_{x,i}^{(L)h_1h_2h_3} = \sum_j d_j(\{p\},\epsilon)\mathrm{MI}_j(\{p\},\epsilon)$$

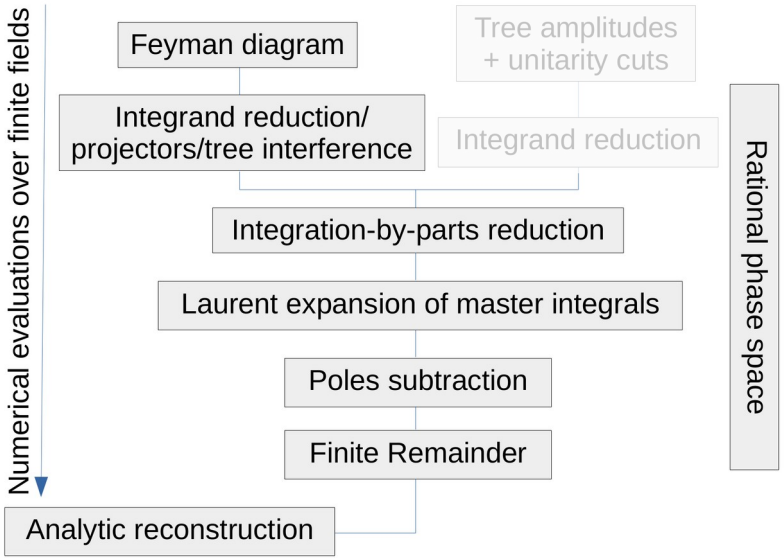
$$\tilde{\mathcal{F}}_{x;i}^{(L)h_3h_4h_5} = \sum_k r_{x;i,k}^{(L)h_3h_4h_5}\,m_k(\vec{f})$$

algebraic

rational

Reconstruction of rational parts

Workflow



Credit: Bayu

QGRAF[Nogueira], FORM[Vermaseren,etal] MATHEMATICA, SPINNEY[Cullen,etal]

finite field framework: FINITEFLOW[Peraro(2019)]

IBP identities generated using LITERED[Lee(2012)] solved numerically in FINITEFLOW using Laporta algorithm[Laporta(2000)]

Mature technology + new optimizations

- Syzygy and module intersection techniques to simplify IBPs [NeatIBP]
- Exploitation of Q-linear relations
- Denominator Ansaetze
- on-the-fly univariate partial fraction

	1	2	3	4
master integral coefficients of mass-renormalised amplitude (full ϵ dependence)	404/393	398/389	411/402	421/411
special function coefficients of finite remainder	314/303	305/296	321/312	326/317
linear relations	291/280	287/278	299/293	304/299
denominator matching #1	291/0	287/0	299/0	304/0
partial fraction decomposition in x_{5123}	44/40	55/51	57/54	58/54
denominator matching #2	44/0	54/0	54/0	56/0
number of sample points (1 prime field)	137076	89624	161482	179838

Algebraic part → special functions

Differential Equations: $\mathrm{d}\vec{\mathrm{MI}} = \mathrm{d}A(\{p\},\epsilon)\vec{\mathrm{MI}}$

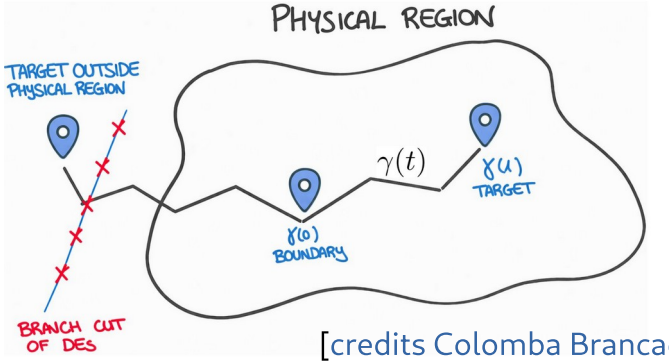
Canonical basis: $\mathrm{d}\vec{\mathrm{MI}} = \epsilon \mathrm{d}\tilde{A}(\{p\})\vec{\mathrm{MI}}$

[Remiddi, 97]
[Gehrmann, Remiddi, 99]
[Henn, 13]

→ not available in ttj case (elliptic integrals):
However, still translates into a system of DEQs for basis functions: $\mathrm{d}G(\vec{d}) = M(\vec{d}) \cdot G(\vec{d})$

$$G(\vec{d}) = \begin{pmatrix} f_i^{(4^*)} \\ f_i^{(4)} \\ \text{weight-3} \\ \text{weight-2} \\ f_i^{(1)} \\ 1 \end{pmatrix}, \qquad M(\vec{d}) = \begin{pmatrix} Y_{4^*,4^*} & 0 & Y_{4^*,3} & Y_{4^*,2} & 0 & 0 \\ 0 & 0 & X_{4,3} & 0 & 0 & 0 \\ 0 & 0 & 0 & X_{3,2} & 0 & 0 \\ 0 & 0 & 0 & 0 & X_{2,1} & 0 \\ 0 & 0 & 0 & 0 & 0 & X_{1,0} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Solve numerically from pre-computed boundary points [AMFlow]



[credits Colomba Brancaccio]

Numerical evaluation of master integrals

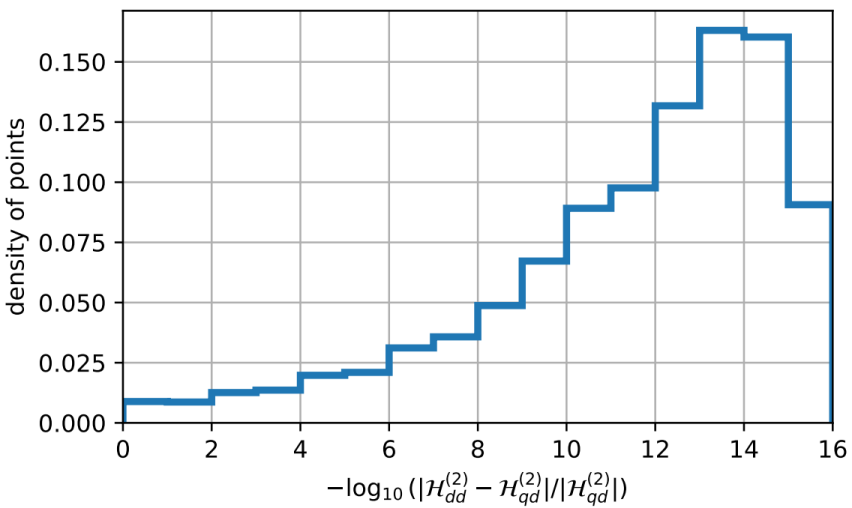
→ Solve DEQ by direct integration
(Bulirsch-Stoer algorithm)
→ Error control based on each function's contribution to squared matrix element

$$\mathcal{Q} \equiv \frac{1}{(4\pi)^4} \frac{2 \sum_{\text{colour}} \sum_{\text{pol.}} \mathcal{R}^{(0)*} \mathcal{R}^{(2)}}{\sum_{\text{colour}} \sum_{\text{pol.}} \left|\mathcal{R}^{(0)}\right|^2}$$

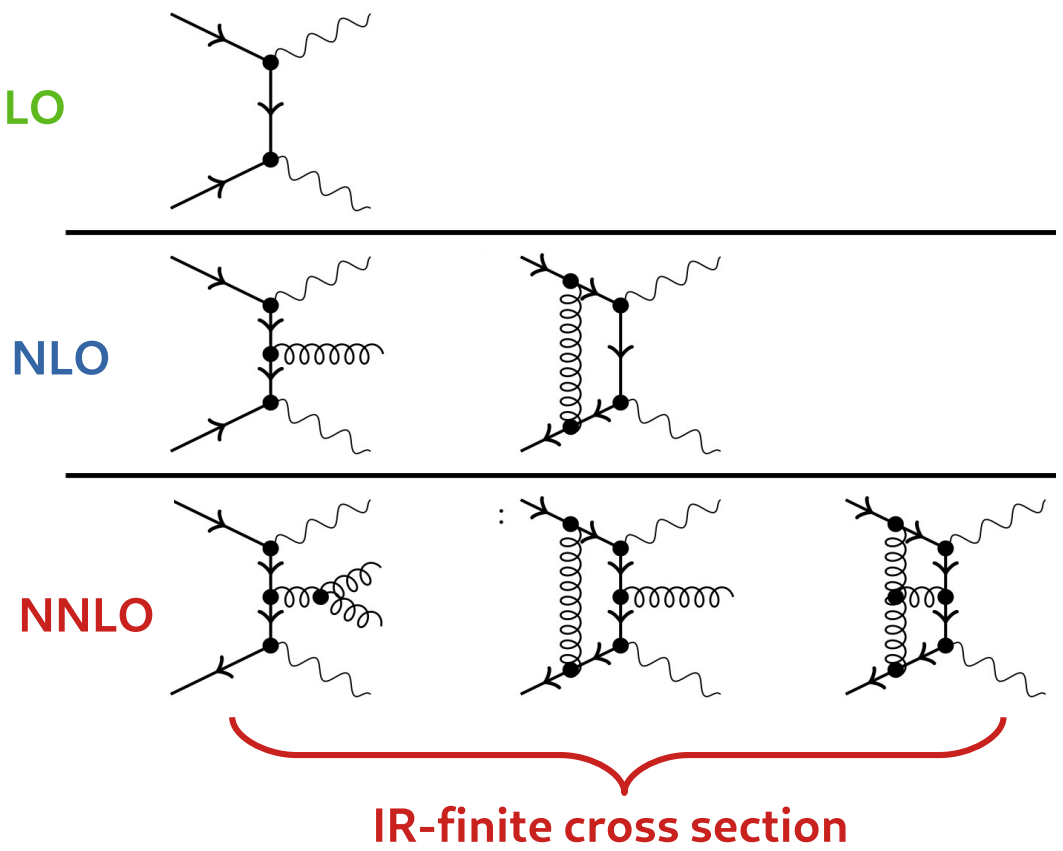
$$\Delta\mathcal{Q}(t) = \left[\sum_{i=1}^{947} \left| \frac{\partial Q}{\partial f_i} \right|^2_{\vec{f}=\vec{f}(t)} \left|\Delta f_i(t)\right|^2 \right]^{1/2}$$

→ directly control numerical error of the final result
→ ready for phenomenology

Channel	Functions [s]	Coefficents [s]	Assembly [s]	total [s]
$gg \to \bar{t}tg$	2.69	30.90	1.58	35.17
$\bar{q}g \to \bar{t}t\bar{q}$	2.16	9.40	0.18	11.74
$qg \to \bar{t}tq$	2.50	9.62	0.21	12.33
$q\bar{q} \to \bar{t}tg$	2.12	9.30	0.18	11.60



NNLO QCD challenges: subtraction



How to achieve **infrared (IR) finite differential** cross sections at NNLO QCD?

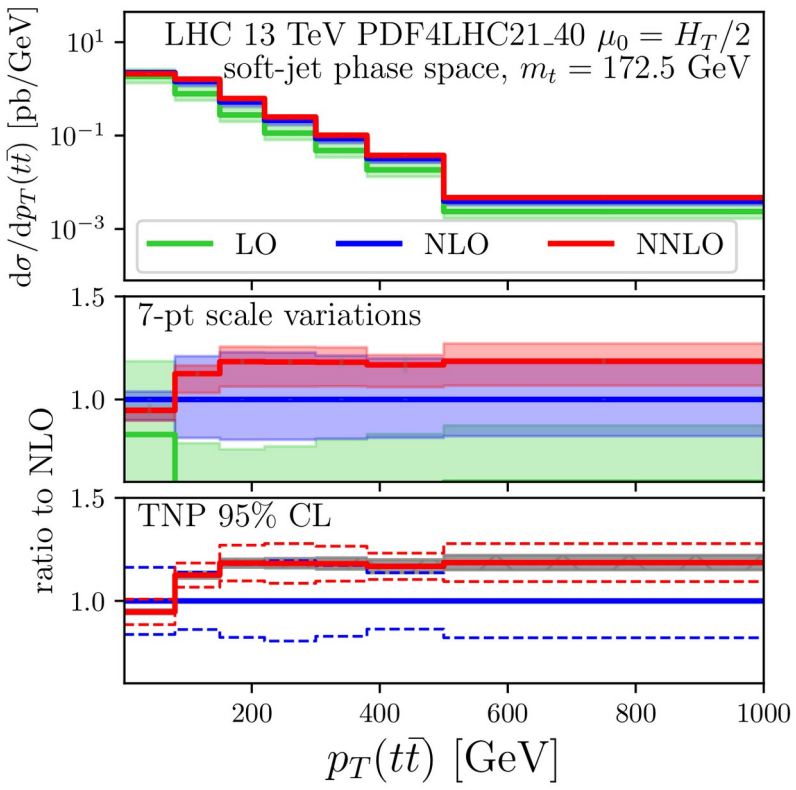
~20 years to solve this problem

qT-slicing [Catani'07], N-jettiness slicing [Gaunt'15/Boughezal'15], Antenna [Gehrmann'05-'08], Colorful [DelDuca'05-'15], Projetction [Cacciari'15], Geometric [Herzog'18], Unsubtraction [Aguilera-Verdugo'19], Nested collinear [Caola'17], Local Analytic [Magnea'18] Sector-improved residue subtraction [Czakon'10-'14,'19]

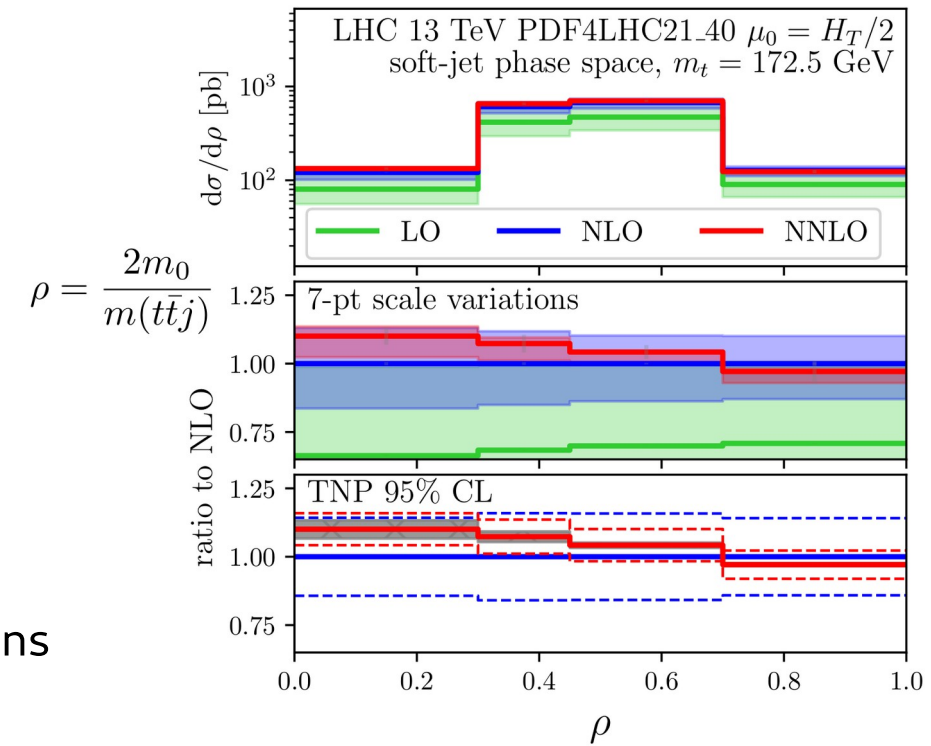
Automation of NNLO QCD subtraction:

- qT-slicing (massive/color singlet) in **MATRIX** & **MCFM** (public)
- Antenna subtraction in **NNLOJet** (public)
- Sector-improved residue subtraction in **Stripper** (private impl.)
- color singlets with nested-collinear in **history** (public)

Differential cross sections

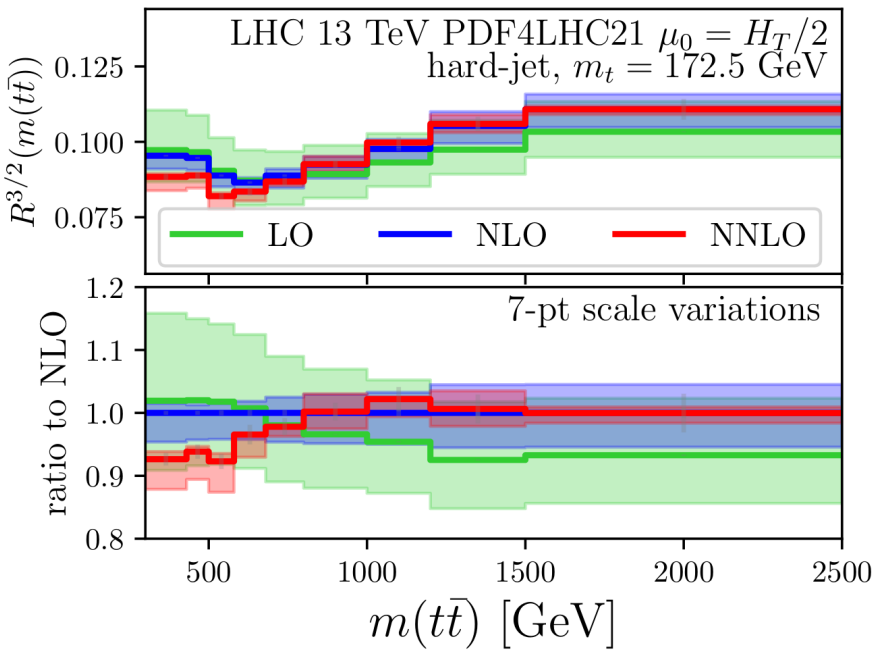


Higher-order QCD corrections to top-quark pair production in association with a jet,
Badger, Becchetti, Brancaccio, Czakon, Hartanto, Poncelet, Zoia
[arxiv:2511.11431]



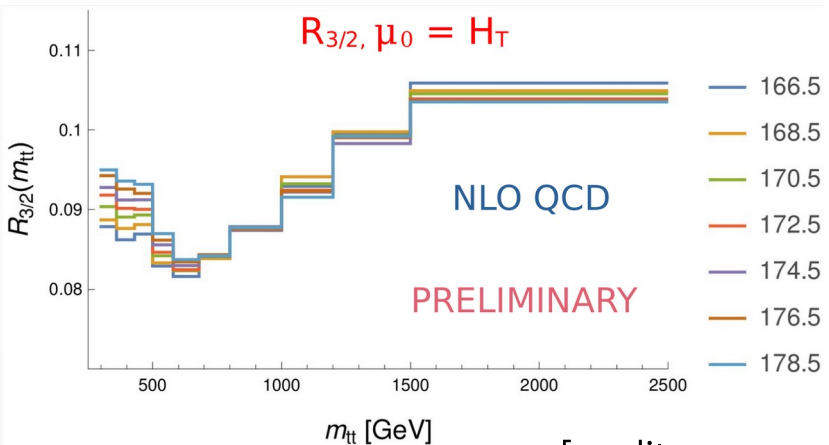
Coloured bands from 7-point scale variations
Dashed bands from 95% CL TNPs

R32 ratios



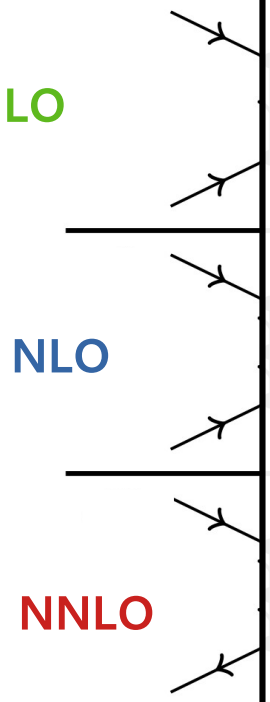
$$R^{3/2}(X) = \frac{\mathrm{d}\sigma^{t\bar{t}j}(X)}{\mathrm{d}\sigma^{t\bar{t}}(X)}$$

→ Sensitivity to α_s & m_t

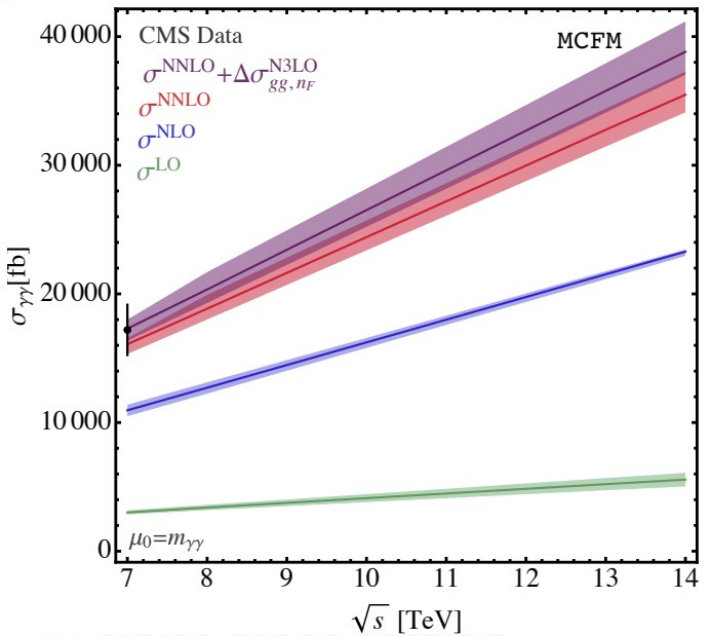


[credits Bayu Hartanto]

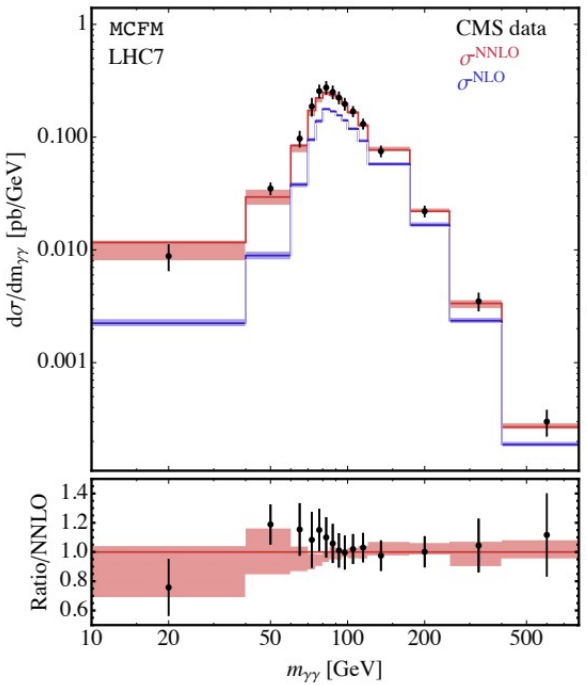
NNLO QCD challenges: large perturbative corrections



NNLO QCD for diphton production [Catani, Cieri, de Florian, Ferrera, Grazzini, 1110.2375] [Campbell, Ellis, Li, Williams, 1603.02663]

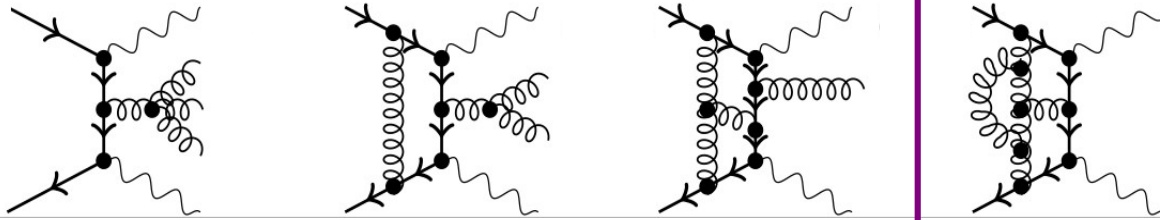


Large NNLO corrections & Loop induced sizeable



Ultimate precision for HL-LHC → towards N3LO QCD

N3LO



Three loop amplitudes:

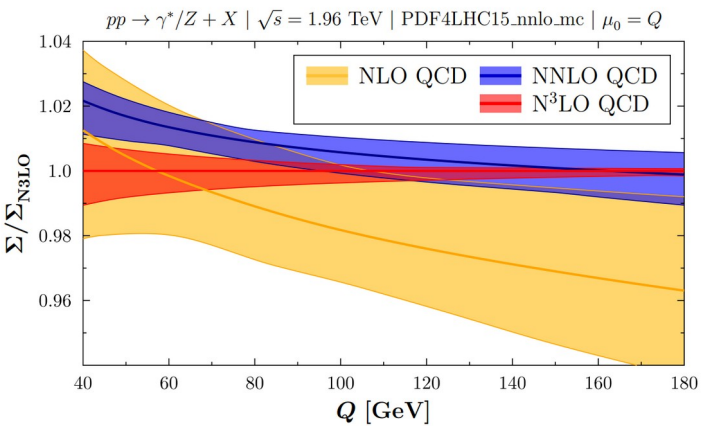
- 2→ 2 partons/photons [Bargiela, Caola, Chakraborty, Gambuti, von Manteuffel, Tancredi '20-'23]
- 2 → 2 with one massive leg:
  - H+jet [Chen, Guan, Mistlberger'25]
  - V+jet [Gehrmann,Jakubčík,Mella, Syrrakos,Tancredi'23]

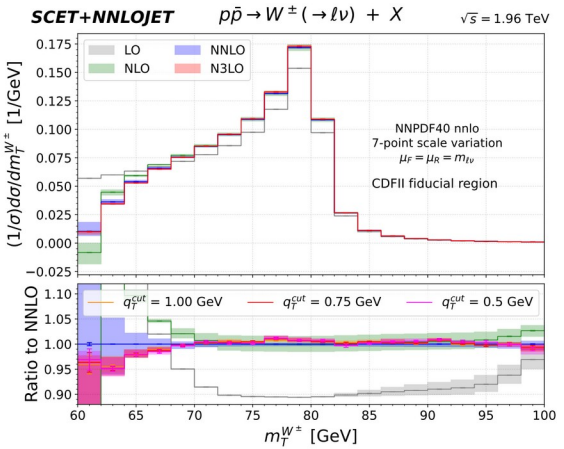
(Semi-)inclusive computations:

- Higgs/DY [Anastasiou, Duhr, Dulat, Furlan, Gehrmann, Herzog, Lazopoulos, Mistlberger, '15-'20]
- VH [Baglio, Duhr, Mistlberger, Szafron '22]
- (Di-)Higgs in VBF [Dreyer Karlberg'16'18]

Fully differential cross sections

- Local e^+e^-→ jets with Antenna [Chen, Jakubčík, Marcoli, Stagnitto'25]
- Projection-to-Born for Higgs in pp [Chen, Gehrmann, Glover, Huss, Mistlberger, Pelloni]
- **Local NNLO + qT slicing** in pp [Chen, Gehrmann, Glover, Huss, Yang, Zhu'21-'22] [Czakon, Eschment, Generet, Poncelet'26]
- Double slicing in DIS [Dong, Fang, Gao,Li,Shao,Zhu,Zhu'26]





N3LO QCD with slicing methods

local subtraction for color singlet+jet cross sections + qT-slicing [Grazzini, Catani'07]
- DY [Chen, Gehrmann, Glover, Huss, Yang, Zhu'21-'22]
- Diphoton [Czakon, Eschment, Generet, Poncelet'26]

local subtraction:
sector-improved
residue subtraction
[Czakon'10' et al 14'19]

$$\sigma^{pp\to\gamma\gamma}_{\mathrm{N^3LO}} = \int_0^{q_T^{\mathrm{cut}}} \mathrm{d}q_T \frac{\mathrm{d}\sigma^{pp\to\gamma\gamma}_{\mathrm{N^3LO}}}{\mathrm{d}q_T} + \int_{q_T^{\mathrm{cut}}}^{\infty} \mathrm{d}q_T \frac{\mathrm{d}\sigma^{pp\to\gamma\gamma j}_{\mathrm{NNLO}}}{\mathrm{d}q_T}$$

computational bottleneck for $q_T^{\mathrm{cut}} \to 0$

QT-factorization →

N3LO TMDs [Ebert, Mistlberger, Vita '20] [Luo, Yang, Zhu, Zhu '20]

$$\frac{\mathrm{d}\sigma^{pp\to\gamma\gamma}}{\mathrm{d}\boldsymbol{q}_T^2\,\mathrm{d}m_{\gamma\gamma}\,\mathrm{d}y} = \sum_{a,b} H_{ab}(m_{\gamma\gamma}) \int \frac{\mathrm{d}^2\boldsymbol{b}_T}{(2\pi)^2} e^{i\boldsymbol{q}_T\cdot\boldsymbol{b}_T} \times \tilde{f}_a(x_1,\boldsymbol{b}_T)\tilde{f}_b(x_2,\boldsymbol{b}_T) + \mathcal{O}(q_T/m_{\gamma\gamma})$$

$$\tilde{f}_i(x,\boldsymbol{b}_T) = \sum_j \int_x^1 \frac{\mathrm{d}z}{z} \tilde{\mathcal{I}}^{\mathrm{TMD}}_{ij}(z,\boldsymbol{b}_T) f_j\left(\frac{x}{z}\right) + \mathcal{O}(\boldsymbol{b}_T\Lambda_{\mathrm{QCD}})$$

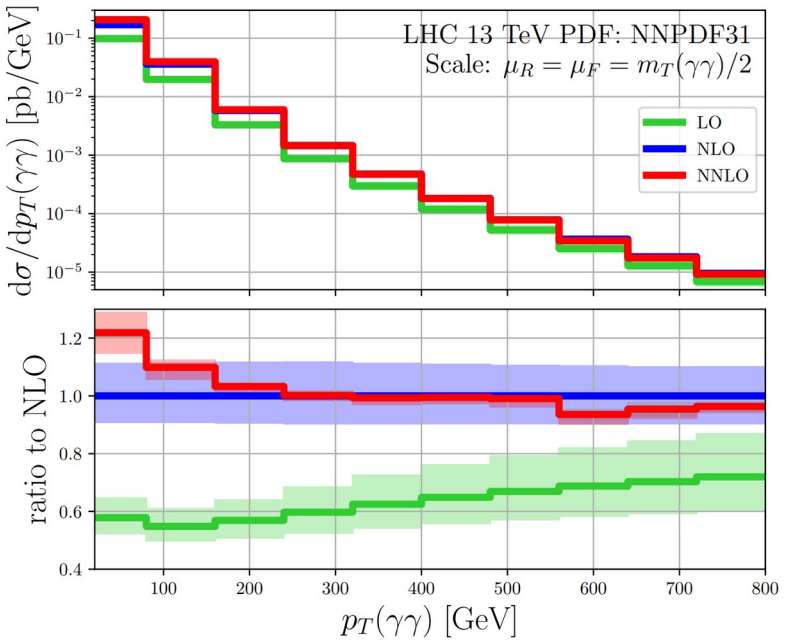
NNLO QCD for photon pair plus jet

Chawdhry, Czakon, Mitov, Poncelet [2103.04319]

with **leading color approx for twoloop**

Agarwal, Buccioni, von Manteuffel,Tancredi [2102.01820]

Chawdhry, Czakon, Mitov Poncelet [2103.04319]

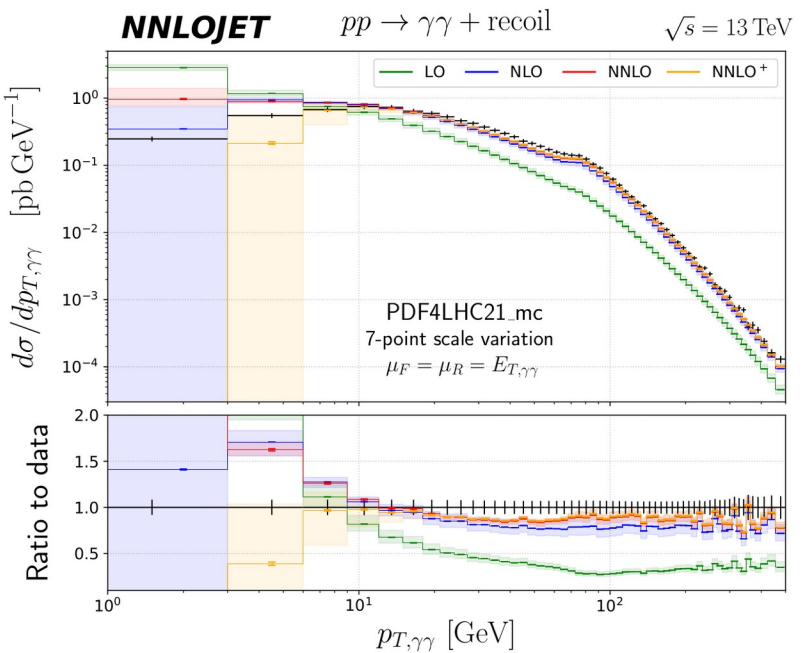


Buccioni, Chen, Feng, Gehrmann, Huss, Marcoli [2501.14021]

Czakon, Poncelet [2512.17591]

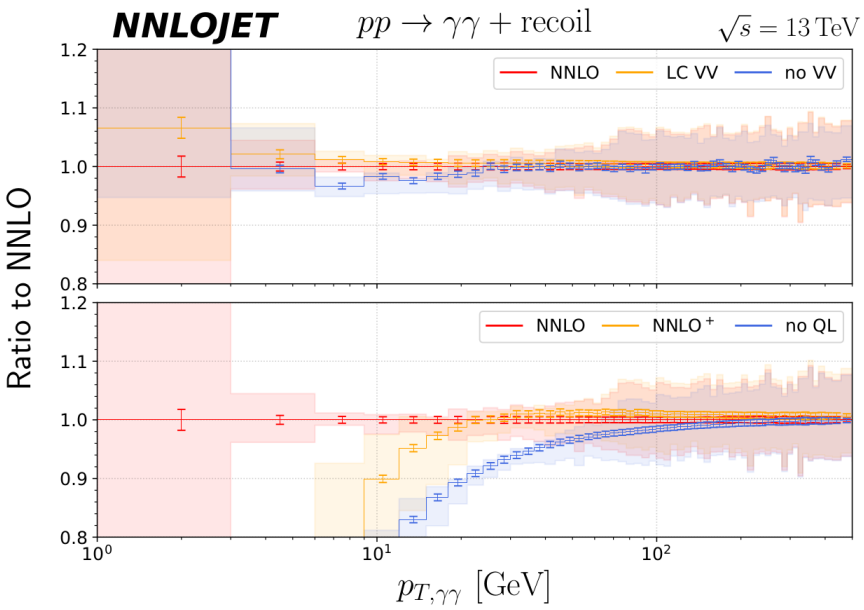
with **complete twoloop**

Agarwal, Buccioni, von Manteuffel,Tancredi [2105.04585]

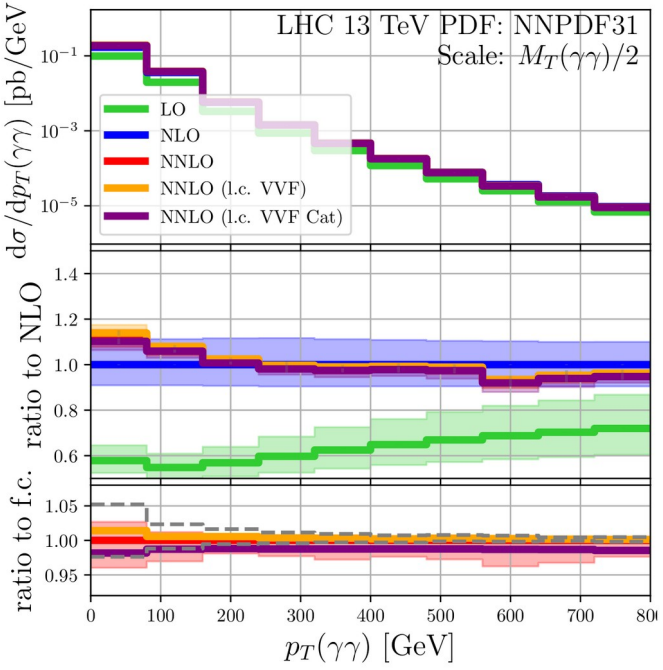


Subleading color effects in di-photon + jet

Buccioni, Chen, Feng, Gehrmann, Huss, Marcoli [2501.14021]



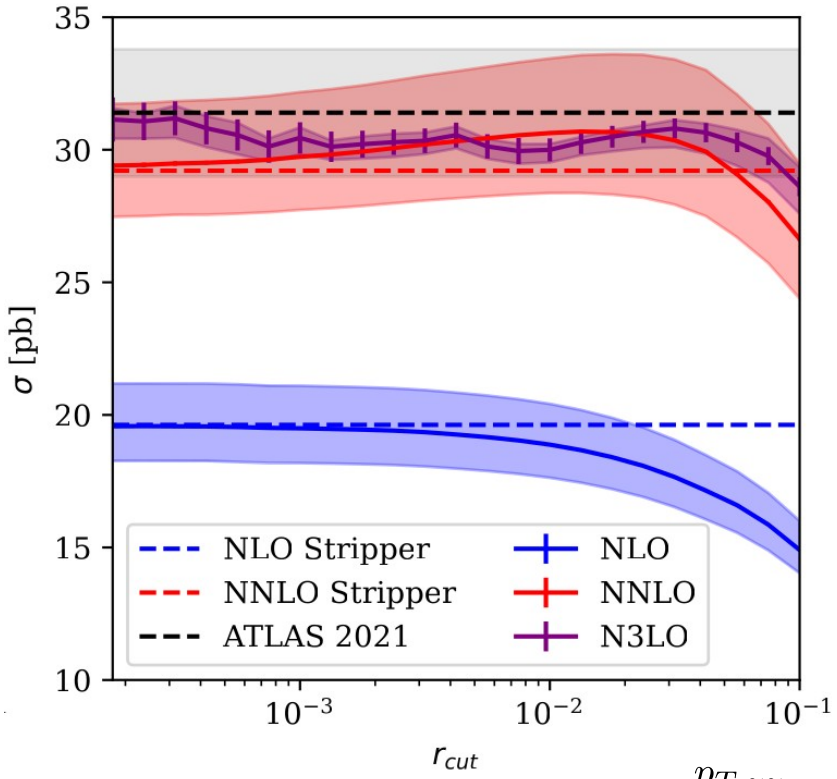
Czakon, Poncelet [2512.17591]



Observation:

two-loop / sub-leading color contribution indeed **small in the bulk**
but can be **important in the corners of phase space**
→ **full color crucial** for applications like qT subtraction

Diphoton fiducial cross section @ N3LO



ATLAS fiducial phase space

$p_{T,\gamma_{1(2)}} \geq 40(30)\ \text{GeV}$ $\qquad$ $\Delta R_{\gamma\gamma} \geq 0.4$

$|\eta_\gamma| < 2.37$ excluding $1.37 < |\eta_\gamma| < 1.52$

hybrid photon isolation:

soft cone: $E_T^{\max}(r) = 0.15\, E_{T,\gamma} \dfrac{1-\cos(r)}{1-\cos(R^s_{\max})},\ r < R^s_{\max}$

hard cone: $E_T^{\max}(r) = 0.09\, E_{T,\gamma},\ r < R^h_{\max}$

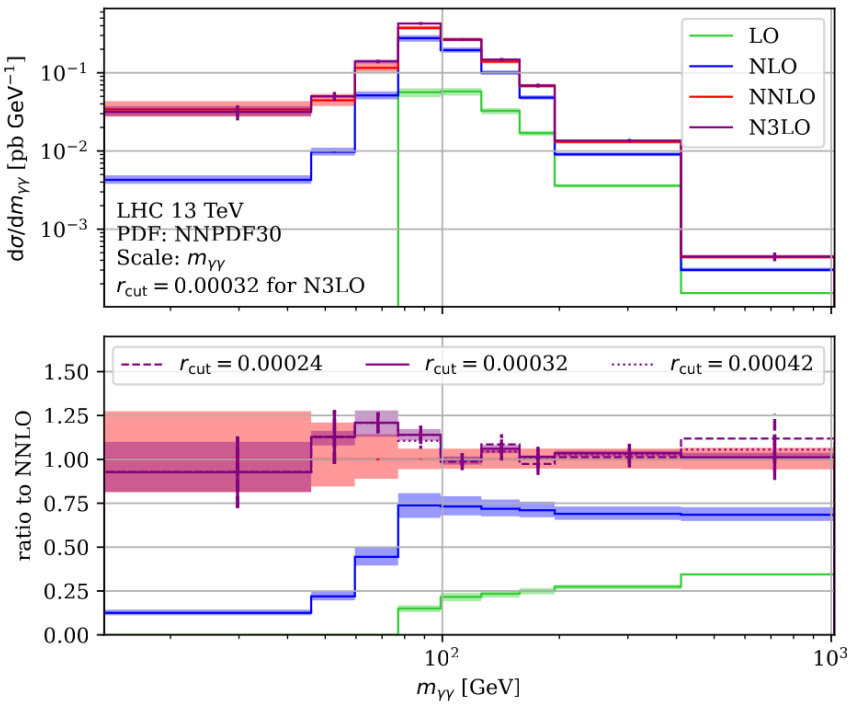
$$\sigma^{pp\to\gamma\gamma}_{\text{N}^3\text{LO}} = 31.2(6)^{+0.5}_{-0.7}\ \text{pb}\,(\pm 0.6\ \text{pb power corrections})$$

for reference: $\sigma^{pp\to\gamma\gamma}_{\text{ATLAS}} = 31.4 \pm 0.1\ (\text{stat.}) \pm 2.4\ (\text{syst.})\ \text{pb}$

as evaluated @ $r_{\text{cut}} = 3.2 \cdot 10^{-4}$

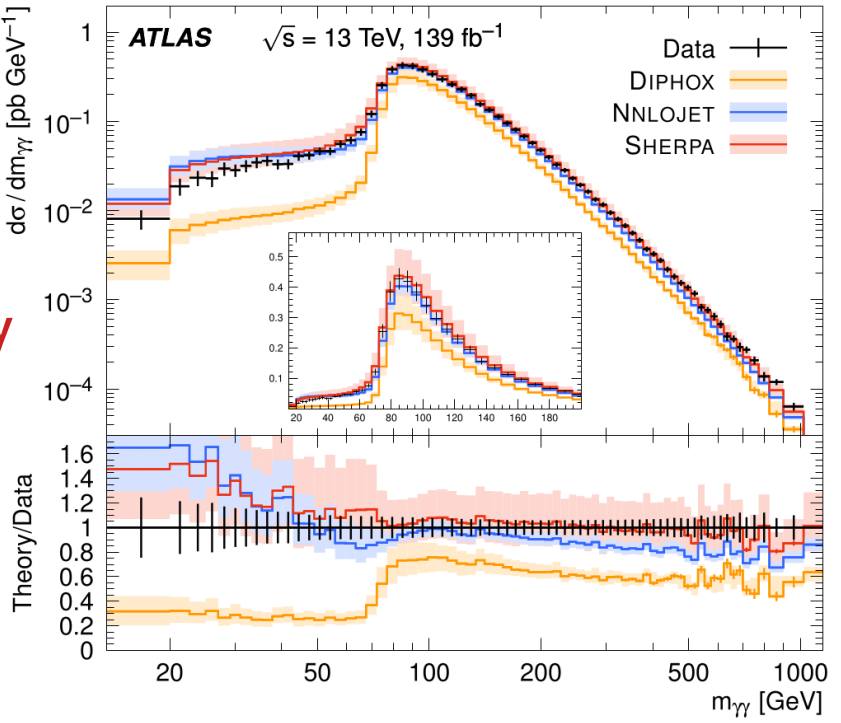
$$r_{\text{cut}} = \frac{p_{T,\gamma\gamma}}{m_{\gamma\gamma}} = 10^{-3} \to p_{T,\gamma\gamma} \approx 0.1\text{GeV}\ (\text{for}\ m_{\gamma\gamma} \approx 100\ \text{GeV}\ @\ \text{PS threshold})$$

Differential distributions



Still a long way to go…

further eff. improvements necessary!



Numerical challenges in slicing computations

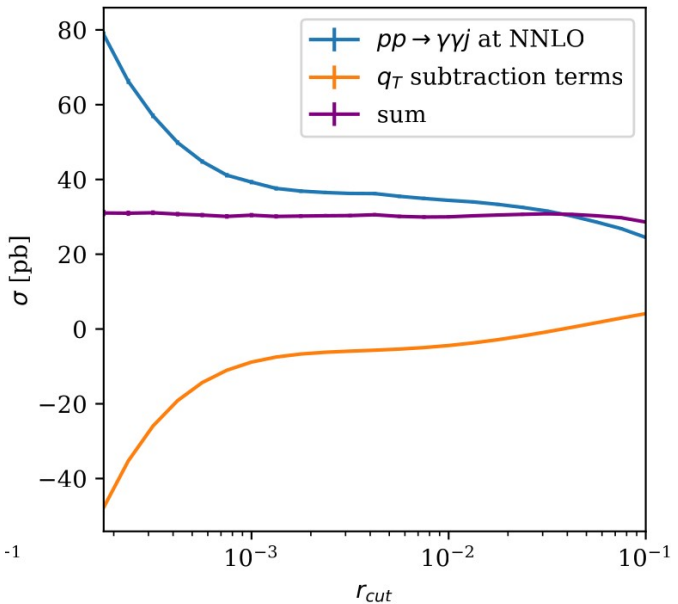
Large power corrections

→ need to go very deep in qT

→ scheme needs to stay stable:

- phase space
- matrix elements
- subtraction terms

} **Many improvements of scheme & implementation!**



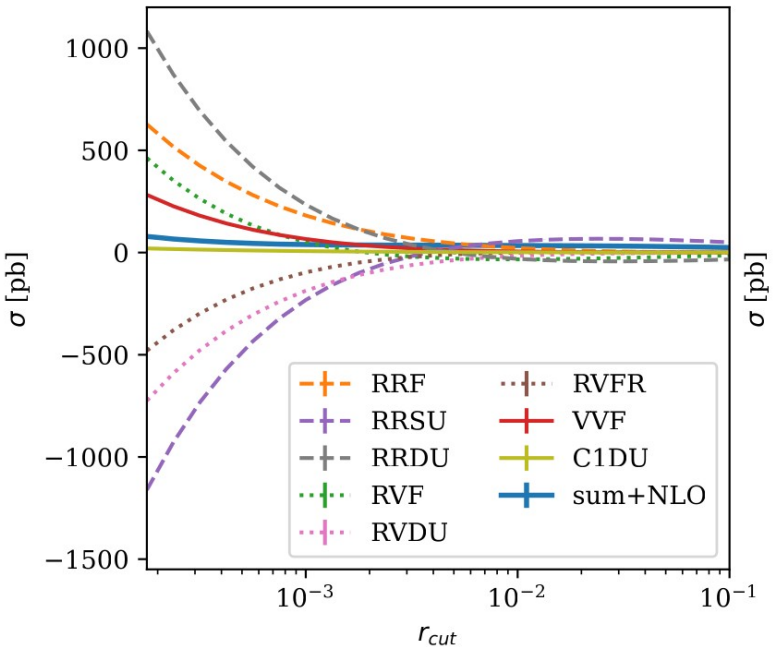
Internal cancellations

between different contributions

→ large negative weight fraction

→ **advance sampling methods**

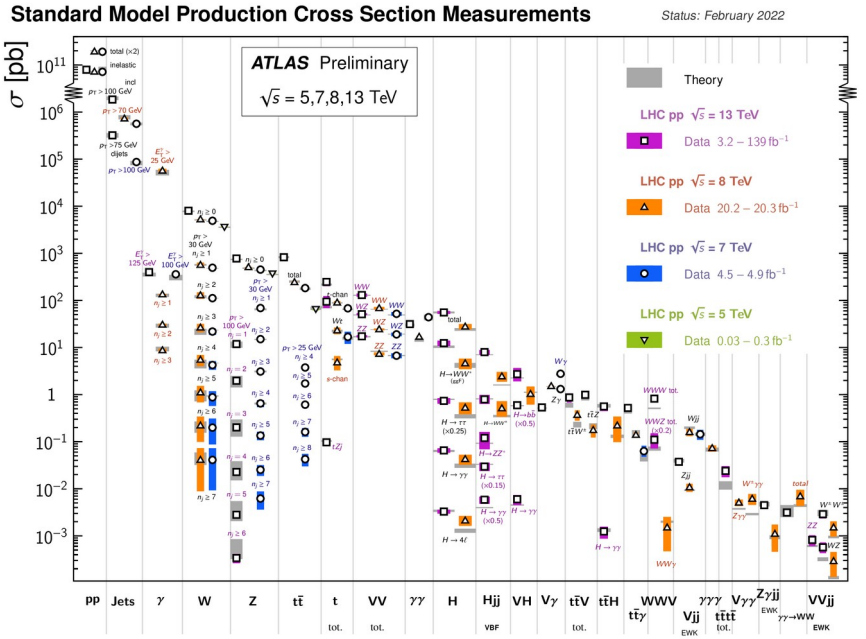
(neural importance sampling)



Take-home messages

- Perturbative QCD is one of the workhorses of SM tests at the LHC

- The community efforts leave **NNLO QCD only limited by multi-scale two-loop amplitudes**
  → current state-of-the-art:
  FC for massless 5-point
  LC for massive 5-point (2 legs + 1 internal)



- **First fully differential N3LO QCD** computation start to appear
  → first **diphoton cross section** (albeit still statistically limited…)
  → motivation for further advancements: local N3LO schemes, improved numerical integration (normalising flows), …

Backup

Sub-leading colour in 2 → 3 processes

Since early 2023 all massless five point amplitudes are known with full color/charge dependence (for photon amplitudes "leading color" ~ "planar")

	Leading colour amplitudes	Full colour amplitudes
$pp \to \gamma\gamma\gamma$	Chawdhry, Czakon, Mitov Poncelet [1911.00479] [2012.13553] Abreu, Page, Pascual, Sotnikov [2010.15834]	Abreu, De Laurentis,Ita, Klinkert, Page, Sotnikov [2305.17056]
$pp \to \gamma\gamma j$	Agarwal, Buccioni, von Manteuffel,Tancredi [2102.01820] Chawdhry, Czakon, Mitov Poncelet [2103.04319]	Agarwal, Buccioni, von Manteuffel,Tancredi [2105.04585]
$pp \to \gamma jj$		Badger, Czakon, Hartanto, Moodie, Peraro, Poncelet, Zoia [2304.06682]
$pp \to jjj$	Abreu, Dormans, Cordero, Ita,Page, Sotnikov [1904.00945] [2102.13609]	Agarwal, Buccioni, Devoto, Gambuti, von Manteuffel, Tancredi [2311.09870] De Laurentis, Ita, Klinkert, Sotnikov [2311.10086] [2311.18752]

all based in pentagon functions: Chicherin, Sotnikov [2009.07803]
(except [1904.00945, 1911.00479] which use: Gehrmann, Henn, Presti [1807.09812])

Machine Learning for efficient Monte Carlo

Variance reduction through importance sampling:

$$\hat{I} = \frac{1}{N}\sum_{i=1}^{N} f(\mathbf{x}_i)\,, \quad \delta\hat{I} = \sqrt{\frac{1}{N-1}\left(\frac{1}{N}\sum_{i=1}^{N} f^2(\mathbf{x}_i) - \hat{I}^2\right)}$$

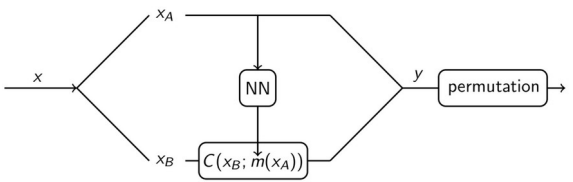
$$I = \int_{\mathbf{H}(\mathbf{x})\in\Omega} \mathrm{d}\mathbf{H}\,\frac{f(\mathbf{x})}{h(\mathbf{x})} \qquad h(\mathbf{x}) = \left|\det\left(\frac{\partial \mathbf{H}(\mathbf{x})}{\partial \mathbf{x}}\right)\right|$$

Task: find $h(\mathbf{x})$ adaptive MC techniques: VEGAS [Lepage'78], Parni [Hameren'14]

Neural importance sampling

Normalising Flows in Sherpa [Bothmann'20], Madgraph [Heimel'22'25],STRIPPER [Jansen'25]

Coupling Layer Flows → series of simple bijections



Continuos Normalising Flows $\dfrac{\mathrm{d}\psi_t(x)}{\mathrm{d}t} = v_t(\psi_t(x))$

Non-positive definite integrands

- Non-definite integrands introduce new challenges

  → cancellation between +/- parts increase the variance

- Consider extreme case: $|f(x)|/h(x) = w = \text{const.}$

  MC estimate:

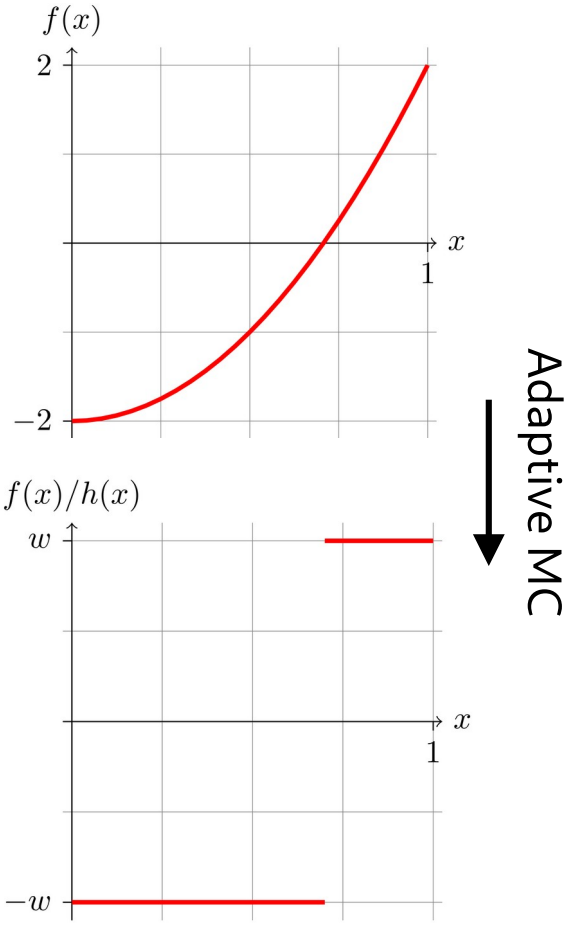
$$\hat{I} = w\frac{N_+ - N_-}{N} \equiv w(2\alpha - 1) \qquad \alpha = N_+/N$$

- Lower bound on variance:

$$\text{Var}(\hat{I}) = w^2 - w^2(2\alpha - 1)^2 = w^2(4\alpha(1-\alpha))$$

→ relative uncertainty: $\frac{\delta\hat{I}}{\hat{I}} = \frac{1}{\sqrt{N-1}}\frac{\sqrt{\alpha(1-\alpha)}}{\alpha - \frac{1}{2}}$

Rephrased: at some point it doesn't matter any more how good your adaptive MC is...



Stratification of signed integrands

$$f(\mathbf{x}) = f_+(\mathbf{x}) + f_-(\mathbf{x})\,, \quad \text{with} \quad f_\pm(\mathbf{x}) = \Theta(\pm f(\mathbf{x}))f(\mathbf{x})$$

$$I = \int_{\mathbf{H}_+(\mathbf{x})\in\Omega} \mathrm{d}\mathbf{H}_+ \, \frac{f_+(\mathbf{x})}{h_+(\mathbf{x})} + \int_{\mathbf{H}_-(\mathbf{x})\in\Omega} \mathrm{d}\mathbf{H}_- \, \frac{f_-(\mathbf{x})}{h_-(\mathbf{x})}$$

"two independent integrals"

$$\hat{I}_{\text{strat}} = \hat{I}_+ + \hat{I}_- = \frac{1}{N_+}\sum_{i=1}^{N_+}\frac{f_+(\mathbf{x}_i)}{h_+(\mathbf{x}_i)} + \frac{1}{N_-}\sum_{i=1}^{N_-}\frac{f_-(\mathbf{x}_i)}{h_-(\mathbf{x}_i)}$$

$$\delta\hat{I}_{\text{strat}} = \sqrt{\frac{1}{N-1}\left[\frac{N}{N_+}\mathrm{Var}(\hat{I}_+) + \frac{N}{N_-}\mathrm{Var}(\hat{I}_-)\right]}$$

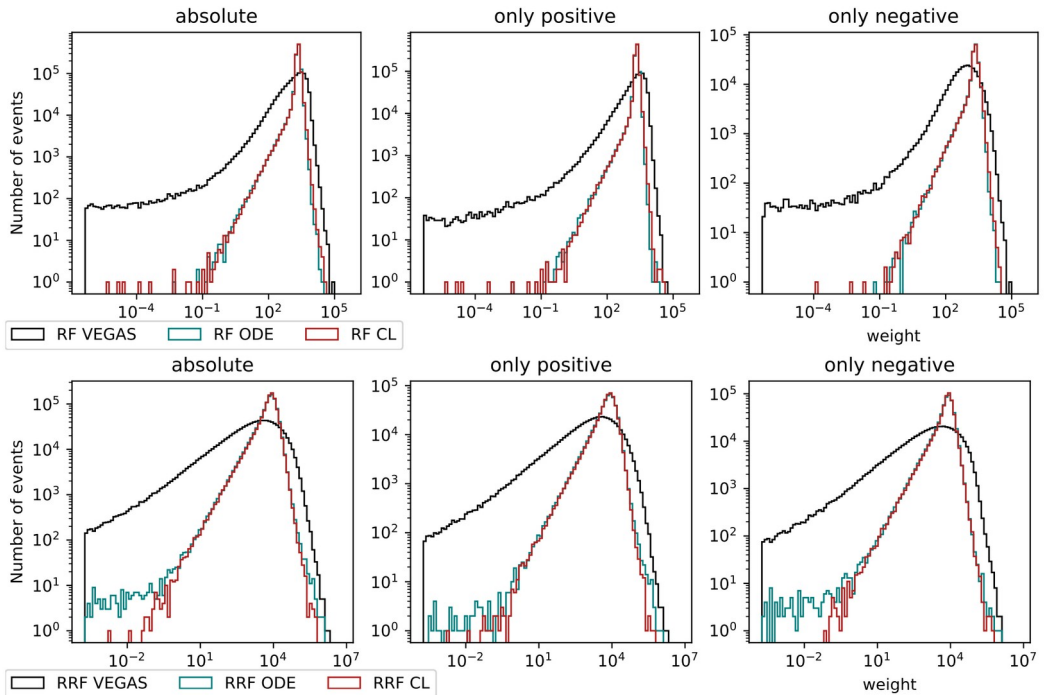
$$\mathrm{Var}(\hat{I}_\pm) = \frac{1}{N_\pm}\sum_{i=1}^{N_\pm}\left(\frac{f_\pm(\mathbf{x}_i)}{h_\pm(\mathbf{x}_i)}\right)^2 - \hat{I}_\pm^2$$

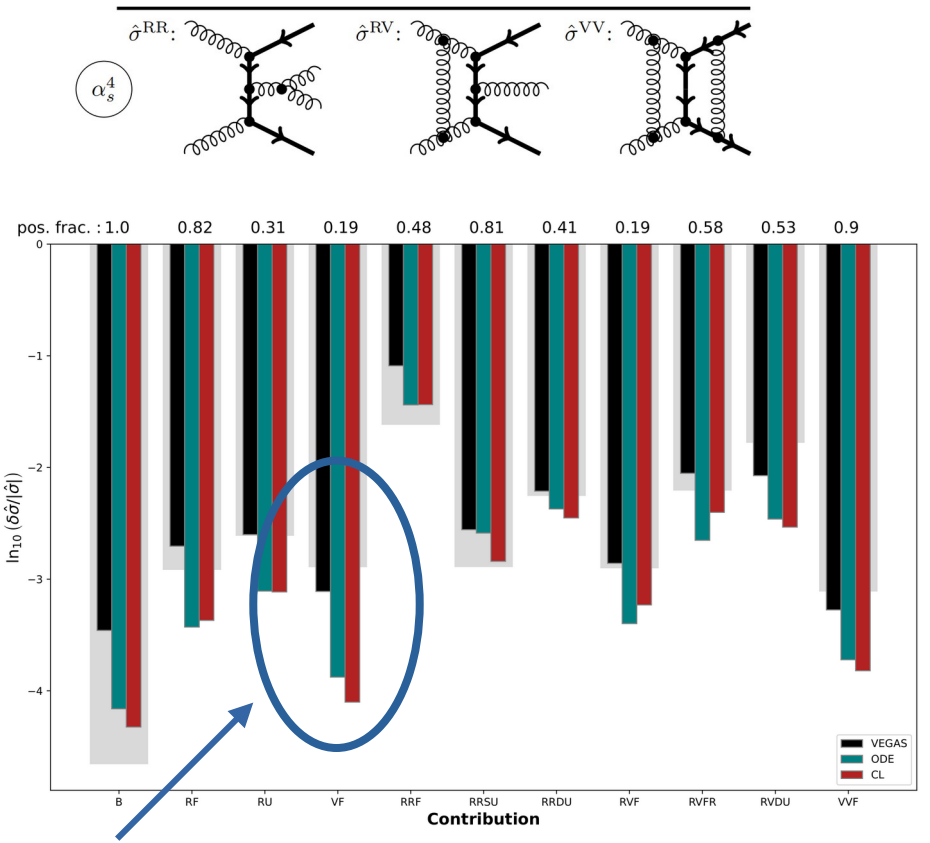
+ The total variance is now bounded by the individual variances

- The mappings are more complicated (need high phase space efficiency)

Normalizing Flows for NNLO QCD predictions

application to gluonic top-pair production:





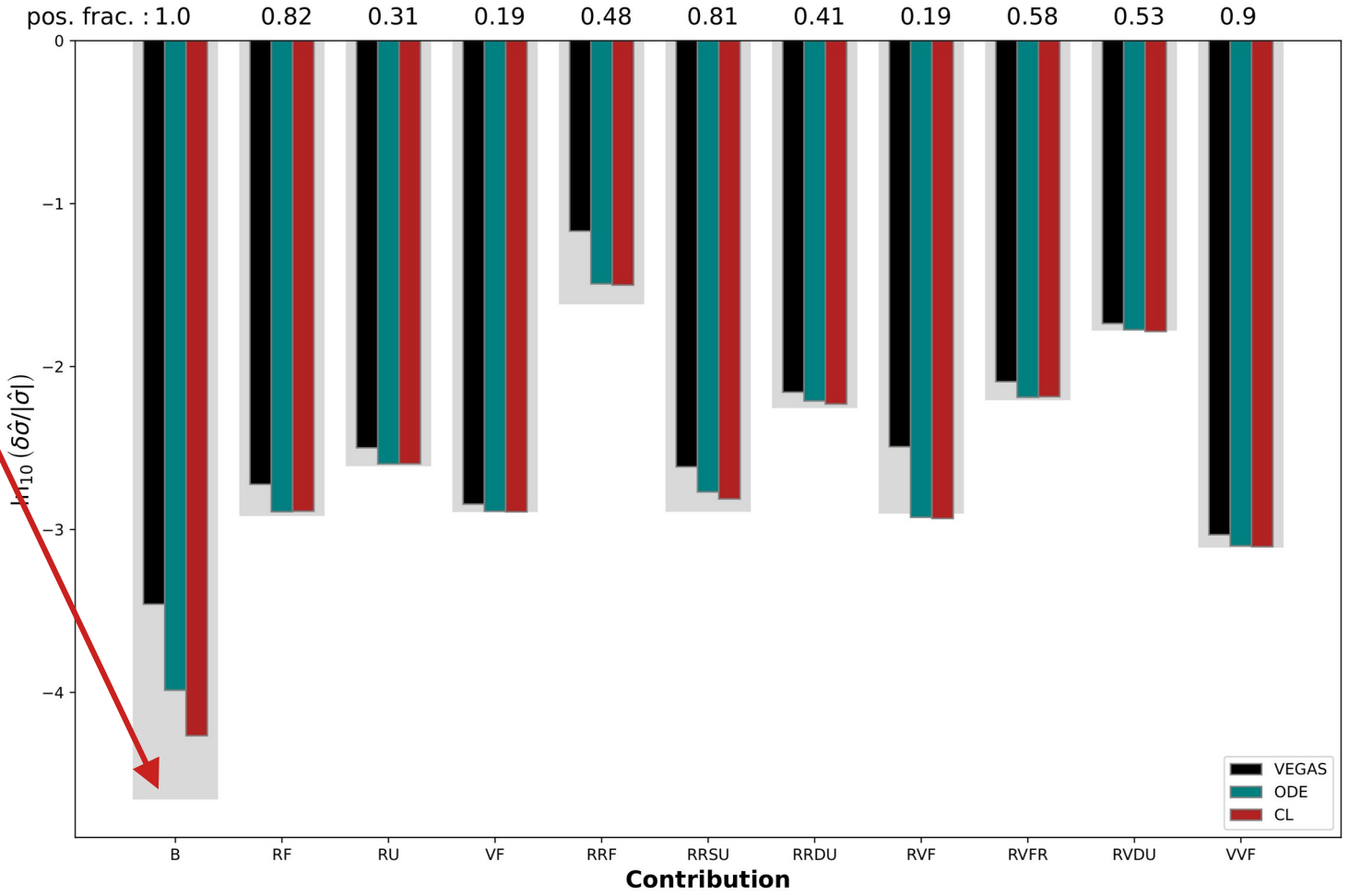
potential for large performance gains

NNLO cross section results → ‘optimal’ uncertainty

Grey boxes
‘Optimal’ uncertainty!

Observation #1:
Saturation of lower bounds both with NF & VEGAS
(for NNLO → NF slightly better)

side remark/interpretation:
1) compliment PS parametrization and/or
2) too simple a problem

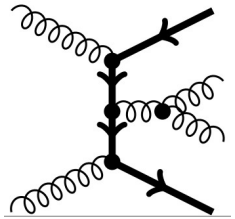


Anatomy of a fixed-order real emission event

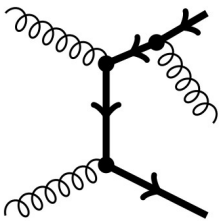
Matrix elements

Kinematics + weight

Observable

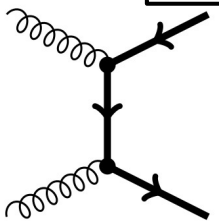


$$\Phi_{n+2},\ w_{n+2} \longrightarrow \mathcal{O}(\Phi_{n+2})$$



$$\otimes\mathcal{S}_1 \longrightarrow \Phi_{n+1},\ w_{n+1} \longrightarrow \mathcal{O}(\Phi_{n+1})$$

generic soft/splitting functions



$$\otimes\mathcal{S}_2 \longrightarrow \Phi_n,\ w_n \longrightarrow \mathcal{O}(\Phi_n)$$

$$w_{n+2} + w_{n+1} + w_n = \text{finite}$$

In all infrared limits

Theory uncertainties

Various sources of theory uncertainties: scales, PDFs, alphaS, …

→ zoom in on two sources on theory uncertainties:

1: **missing-higher-order-uncertainties (MHOU)**

2: **missing sub-leading colour contributions**

Theory uncertainties from scale variations

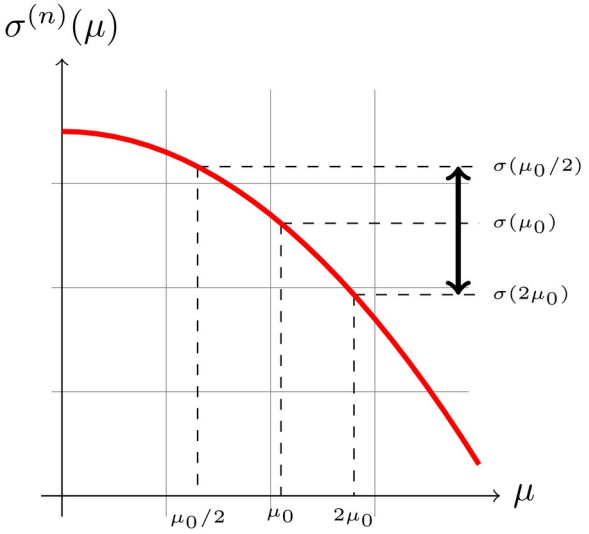
Lets focus on QCD as an example: $\alpha = \alpha_s(\mu_0)$

$$\mathrm{d}\sigma = \mathrm{d}\sigma^{(0)} + \alpha_s \mathrm{d}\sigma^{(1)} + \alpha_s^2 \mathrm{d}\sigma^{(2)} + \ldots \xrightarrow{\text{RGE}} \mu \frac{\mathrm{d}\sigma^{(n)}}{\mathrm{d}\mu} = \mathcal{O}\left(\alpha_s^{(n_0+n+1)}\right)$$

Of same order as the next dominant term → exploiting this to estimate size of $\mathrm{d}\sigma^{(n+1)}$

Scale variation prescription (ad-hoc and heuristic choice)

- choose 'sensible' μ_0
  - → principle of fasted apparent convergence: $\sigma^{(n)}(\mu_{\mathrm{FAC}}) = 0$
  - → principle of minimal sensitivity: $\mu \frac{\mathrm{d}\sigma(\mu)}{\mathrm{d}\mu}\Big|_{\mu=\mu_{\mathrm{PMC}}} = 0$
  - → ...
- vary with a factor (typically 2)
- take envelope as uncertainty



Short comings of scale variations

- not always reliable … however in most cases issues are understood/expected:
  new channels, phase space constraints, etc. → often we can design workarounds
- however, some issues are more fundamental:
  - → how to choose the central scale? → not a physical parameter, no 'true' value
    (Principle of fasted apparent convergence, principle of minimal sensitivity,…)
  - → how to propagate the estimated uncertainty, no statistical interpretation!
  - → what about correlations? Based on 'fixed form' of the lower orders and RGE.

(At the moment) two alternative approaches under investigation:

"Bayesian"

[Cacciari,Houdeau 1105.5152]
[Bonvini 2006.16293]
[Duhr, Huss, Mazeliauskas, Szafron 2106.04585]

"Theory Nuisance Parameter"

[Tackmann 2411.18606]
→ W mass extraction: [CMS 2412.13872]
[Cal,Lim, Scott,Tackmann Waalewijn 2408.13301]
[Lim, Poncelet, 2412.14910]
[Cridge,Marinelli,Tackmann,2506.13874]

TNP approach for fixed-order computations

[Lim, Poncelet, 2412.14910]

$$\begin{aligned}\mathrm{d}\sigma &= \alpha_s^n N_c^m \,\mathrm{d}\bar{\sigma}^{(0)} + \alpha_s^{n+1} N_c^{m+1} \,\mathrm{d}\bar{\sigma}^{(1)} + \alpha_s^{n+2} N_c^{m+2} \,\mathrm{d}\bar{\sigma}^{(2)} + \dots \\ &= \alpha_s^n N_c^m \,\mathrm{d}\bar{\sigma}^{(0)} \left[1 + \alpha_s N_c \left(\frac{\mathrm{d}\bar{\sigma}^{(1)}}{\mathrm{d}\bar{\sigma}^{(0)}}\right) + \alpha_s^2 N_c^2 \left(\frac{\mathrm{d}\bar{\sigma}^{(2)}}{\mathrm{d}\bar{\sigma}^{(0)}}\right) + \dots\right]\end{aligned}$$

Observation, i.e. “expert knowledge”: $\dfrac{\mathrm{d}\bar{\sigma}^{(i)}}{\mathrm{d}\bar{\sigma}^{(0)}} \sim \mathcal{O}(1)$

Use some knowledge about lower orders but introduce parametric dependence:

$$\frac{\mathrm{d}\bar{\sigma}_{\mathrm{TNP}}^{(N+1)}}{\mathrm{d}\bar{\sigma}^{(0)}} = \sum_{j=1}^{N} f_k^{(j)}\left(\vec{\theta}, x\right)\left(\frac{\mathrm{d}\bar{\sigma}^{(j)}}{\mathrm{d}\bar{\sigma}^{(0)}}\right)$$

x → mapped kinematic variable

Approximation of original TNP philosophy
→ there is only $f_i(\hat{\theta}) \approx \hat{f}_i$

Chebyshev: $f_k^C\left(\vec{\theta}, x\right) = \dfrac{1}{2}\sum_{i=0}^{k} \theta_i T_i(x) \qquad x \in [-1, 1]$